

— FINAL-YEAR PROJECT

# An Agentic Multi-LLM Answer-Validation System

Can a system of language models decide, on its own,  
whether an answer is trustworthy — with no human  
answer key to check against?

**Generators** LLaMA 3 · Mistral   **Verifier** phi3   **Judge** qwen3.5:9b   **Dataset** 5,000 questions

# A model can't grade its own homework

- 1 Ask a language model a hard question and it answers fluently — **whether or not it's correct**, and it can't reliably tell you which.
- 2 Having a second model "check" the first fails quietly — checkers **share the same biases** and rubber-stamp plausible text.
- 3 In this domain there is **no ground-truth answer key** — so accuracy can't be measured directly. Every design choice must be justified another way.

# Trust through independent corroboration

Instead of one model checking another, use an ensemble that cross-examines itself — and gate the output on how much that ensemble agrees.

**Core principle:** two models generate, two more evaluate blindly, three external checks corroborate, and one confidence score decides whether to ship the answer or try again.

# Six stages, one confidence score

STAGE 1-2

## Generate

LLaMA 3 + Mistral,  
independently

STAGE 3

## Verify

phi3 scores both, blind

STAGE 5

## Judge

qwen3.5:9b scores + picks  
winner

STAGE 4

## Corroborate

Wikipedia · Wikidata ·  
agreement

STAGE 6

## Gate + Combine

score  $\geq 0.75$  or regenerate

**The key control:** a per-question coin flip randomizes answer order, and evaluators never see which model wrote which — so position and brand can't leak into a score.

# Every choice earned its place

With no correctness labels, validity came from two families of label-free test — both of which can *fail*, which is what makes them real.

## DISCRIMINATION

### Can it tell bad from good?

A usable evaluator must score a deliberately broken answer far below a solid one. This test alone rejected two judge models.

## CONSISTENCY & VALIDITY

### Do the signals cohere?

Signals should predict each other, and a low confidence score should genuinely predict that a retry improves the answer.

# Judge chosen on discrimination, not averages

| JUDGE MODEL     | MEAN / 80 | FLAT-RATE | VERDICT          |
|-----------------|-----------|-----------|------------------|
| qwen2.5         | 71.9      | —         | Ceiling bias     |
| CompassJudger-1 | 69.3      | 57.5%     | No real analysis |
| qwen3.5:9b      | 65.5      | 0.0%      | Adopted          |

CompassJudger-1 gave all eight rubric criteria the **identical score** on 57.5% of answers — a fluent number with no reasoning behind it. Similar averages hid a decisive difference visible only per-criterion.

# Position bias was hiding as quality

Early on, one model was always shown first and prompts named the brands. Once evaluation was made blind and order-randomized, the apparent gap collapsed:

BY RAW POSITION (ARTEFACT)

**35% · 65%**

A vs B — inflated by the ordering

BY TRUE MODEL IDENTITY

**45% · 55%**

LLaMA 3 vs Mistral — the real, small gap

Separating the two proved most of the "gap" was an artefact — and quantified the modest part that was genuine.

# The 0.75 threshold, proven three ways

## METHOD

## RESULT

Two-proportion z-test

$p = 0.011$  — the drop-off is real

ROC / Youden's J

optimal 0.70–0.74, AUC 0.63

**2,000-sample bootstrap**

**95% CI [0.69, 0.75]**

**Solved by design:** a control group gave a random slice of already-passing answers a retry too — the only way to get unbiased data above the threshold.

# Three measurement bugs — caught, not shipped

- 1 A model's **self-reported total** didn't match its own scores → now always recomputed from the parts.
- 2 A "second-opinion" signal was **secretly derived from the judge**, inflating results → excluded from the search.
- 3 A threshold analysis **under-counted part of the data** after a sampling rate changed → re-weighted correctly.

Each was found by a result looking *too clean* and being checked — and corrected before it became a conclusion.

# What the evidence doesn't settle

The system prefers Mistral (55%). Is that real quality, or a judge quirk?

THE JUDGE SAYS

**Mistral is more factual**

Higher on correctness, faithfulness, hallucination-freedom.

THE VERIFIER SAYS

**No difference at all**

Same answers, zero gap on any criterion — and it's proven able to spot real gaps.

Reported as **open**, with the evidence for both sides — not forced to a tidy conclusion.

— 10 · DELIVERED

# The project at a glance

**5,000**

questions, one consistent  
regime

**8,432**

attempts, every signal  
logged

**6**

pipeline stages, fully  
instrumented

**35**

dated research-log entries

A working pipeline, a bias-corrected dataset, a statistically calibrated confidence gate, and a transparent record of how every decision was justified.

— IN ONE LINE

The contribution isn't a pipeline that runs — it's a **method for trusting it** where no ground truth exists.

Thank you    Questions welcome